CREDIT QUALITY & CUSTOMER RISK PROFILING REPORT

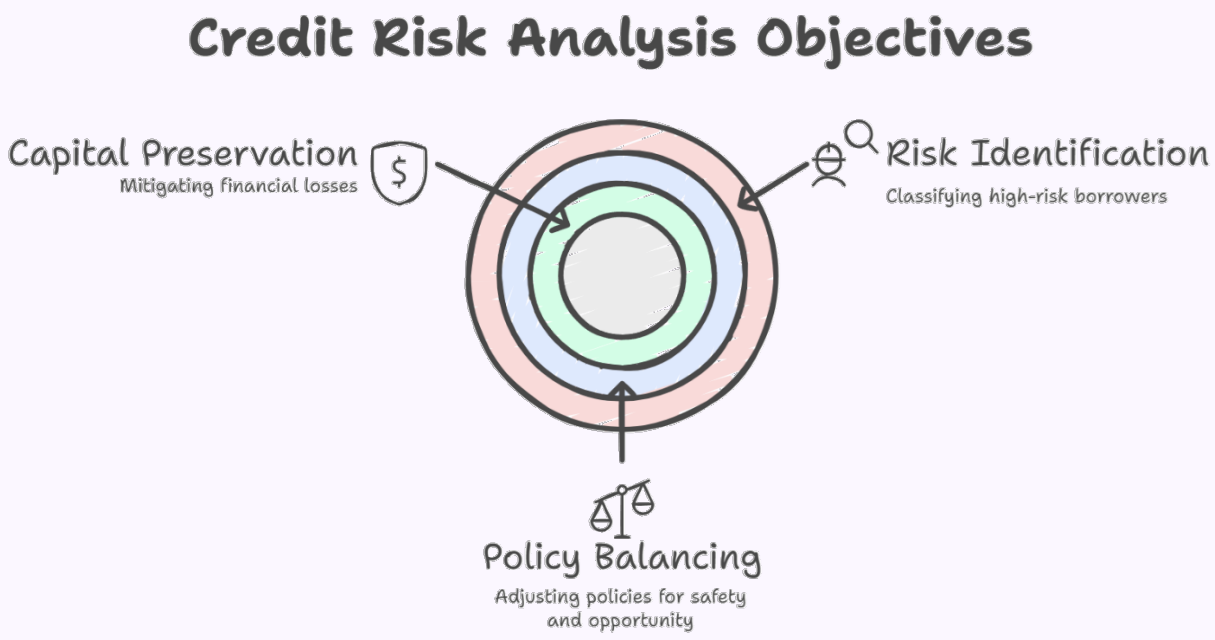
I. EXECUTIVE OVERVIEW

This report analyzes the loan portfolio of **Nova Bank** across its global markets, including the USA, UK, and Canada. The primary mission is to **optimize lending decisions**: expanding credit access for reliable borrowers while accurately filtering out high-risk profiles to prevent financial loss.

By analyzing historical borrower data, this project identifies key risk drivers and introduces a high-accuracy predictive model to make credit decisions more reliable and data-driven.

Strategic Focus:

- **Risk Identification:** Identifying specific borrower profiles and behaviors that lead to default.
- **Capital Preservation:** Mitigating financial losses by highlighting high-risk segments.
- **Policy Balancing:** Refining lending thresholds and interest rate policies to ensure long-term profitability.

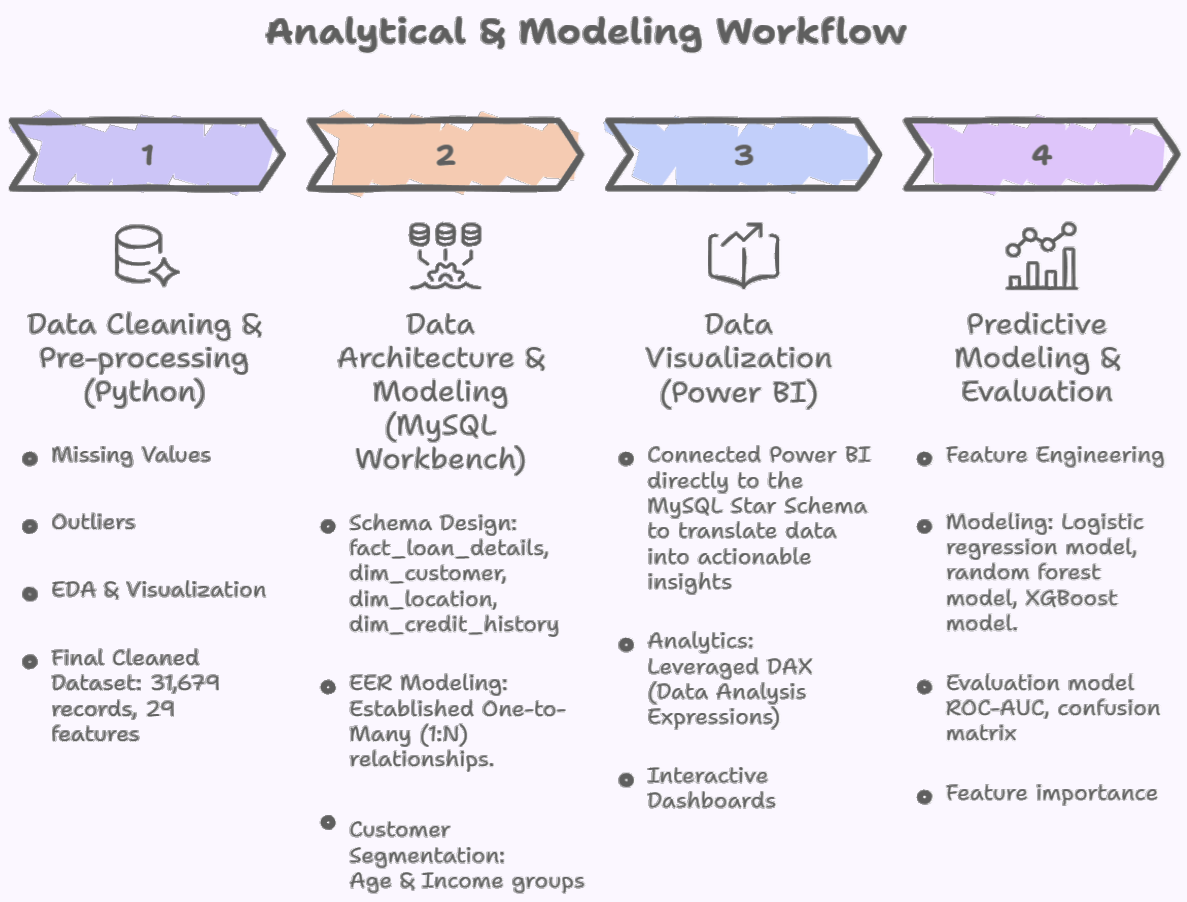


II. DETAILS & METHODOLOGY

1. Dataset Information (Raw Dataset)

- **Size:** 32.581 records | 29 features.
- **Target Variable:** loan_status (0: Non-Default | 1: Default).
- **Data Dictionaries:**
  - **Customer Profile:** person_age, gender, marital_status, education_level
  - **Financial information:** person_income, person_emp_length, employment_type, person_home_owership
  - **Loan details:** loan_amnt, loan_term_months, loan_int_rate, loan_grade,loan_intent, loan_status, open_accounts, other_debt
  - **Ratios:** loan_percent_income, loan_to_income_ratio, debt_to_income_ratio, cb_person_cred_hist_length,cb_person_default_on_file, credit_utilization_ratio,
  - past_delinquencies
  - **Geographic information:** country, state, city, city_latitude , city_longitude
  - **ID:** client_ID

2. Analytical & Modeling Workflow



a) Data Cleaning & Pre-processing (Python)

The project began with a raw dataset comprising 32,581 records and 29 columns. Data cleaning and exploratory analysis were prioritized to refine the features and establish a solid foundation for the XGBoost modeling phase.

- **Missing Values:** Removed records with missing employment length; imputed missing interest rates using the median values grouped by Grade and Purpose.
- **Outliers:** Filtered unrealistic data points (Age > 100 and cases where employment length was inconsistent with the borrower's age).
- **EDA & Visualization:** Utilized Boxplots and Scatter plots to analyze distributions, skewness, and feature correlations.
- **Final Cleaned Dataset:** 31,679 records, 29 features

b) Data Architecture & Modeling (MySQL Workbench)

To ensure high-performance analytics, the cleaned dataset was structured into a Star Schema:

- **Schema Design:**
  - fact_loan_details: Stores quantitative metrics (loan amount, interest, status).
  - dim_customer: Borrower demographics.
  - dim_location: Geographical attributes.
  - dim_credit_history: Historical credit performance.
- **EER Modeling:** Established One-to-Many (1:N) relationships between Dimension tables and the Fact table to ensure referential integrity.
- **Dynamic Segmentation (SQL Views):** Age & Income Groups: Engineered custom SQL Views to categorize borrowers based on statistical percentiles.

c) Data Visualization (Power BI)

Connected Power BI directly to the MySQL Star Schema to translate data into actionable insights:

- **Interactive Dashboards:** Developed dynamic reports to track the $306M portfolio, visualizing risk hotspots across USA, UK, and Canada.
- **Analytics:** Leveraged DAX (Data Analysis Expressions), including a 21.5% Default Rate and 24.6% NPL Ratio based on the cleaned 31,679 records.

d) Predictive Modeling & Evaluation

- **Feature Engineering:** Categorical encoding and feature scaling performed on the structured data.
- **Modeling:** Trained XGBoost model to handle class imbalance (~21.5% default rate). The XGBoost model delivered exceptional predictive performance.
- **ROC-AUC: 0**.93 – Demonstrated superior capability in distinguishing between "Non-Default" and "Default" profiles.
- **Recall:** 0.75 – Proactively identified 75% of actual defaults, allowing the bank to mitigate risks at the pre-disbursement stage

III. KEY INSIGHTS

1. Portfolio Quality & Geographic Distribution

Analysis based on 31,679 records ($306M) across USA, UK, and Canada.

- **KPI Benchmarks :**
  - **Default Rate (Volume-based**): 21.5% ~ 6.8K Bad loan applications
  - **NPL Ratio** (Value-based): 24.6% ~ $75M Bad loan amount
- **Uniform Market Performance:** Default rates are consistent across the USA, UK, and Canada. This indicates that credit risk is fundamentally driven by borrower financial profiles rather than geographical location.
- **Safety Zone (Low risk):**
  - Grade A (9.6% Default) and Venture loans (14.7% Default) represent the lowest risk segments.
  - Grade B, A act as the portfolio's anchor, securing $192M in total lending.
- **Extreme Risk Zone (High Risk):** Risk spikes in lower grades.
  - Grade G increases default probability. Grade D is the most critical concern as it contributes the most to the total portfolio risk.
  - Purpose: Medical and Debt Consolidation loans within Grades D-E-F-G suffer near-total loss rates (90%–100%).

2. High-Risk Personas & Behavioral Patterns

- **Low-to-Medium Income + Youth Segment:**
  - **Market Dominance:** Gen Z and Young Adults (ages 20-30) are the primary customer base, ~76% of the total portfolio.
  - **Concentrated Risk**: this segment is the single largest source of financial loss. Specifically, 4.8K borrowers under 30 (income <$80k) contributed $49M in bad debt, which accounts for a about ~65% of the total portfolio's bad debt.
  - **Stability Red Flags**: Risk peaks significantly among Renters and Mortgage holders with working experience less than 2 years. When Interest Rates > 15%, the Default Rate and NPL ratio ~ to 87%.
- **Credit Precedent:** Customers with a prior delinquency record have a 37.6% default rate, twice as high as those with a clean history.

3. The Interest & DTI Gap (Management Insight)

- **The 15% Interest Threshold:** Interest rates above 15% drive default rates to 57.9%.
- **The Leverage Trap:** The combination of **DTI > 50% and Interest > 15%** resulted in an **NPL ratio of 78.8%** ($9M in bad debt out of $11M disbursed). This proves that high interest cannot compensate for extreme financial leverage.

IV. RECOMMENDATIONS

1. Underwriting Policy Adjustments

- **AI-Powered Preliminary Assessment:** Integrate XGBoost models for real-time default forecasting. Automatically flag High-Risk Grades (D-G) and risky loan purposes (Medical/Debt Consolidation) at the point of application.
- **Strict DTI & Interest Control:** Prohibit interest rates **>15%** for borrowers with **DTI >50%**. Data proves this specific overlap leads to **78.8% NPL ratio**.
- **Credit Screening:**
  - Mandate high-liquidity collateral for prior delinquencies.
  - Limit approval loan amount for Medical/Debt Consolidation in Grades D-G.

2. Targeting Strategy

- **Expand the "Safe Zone":** Aggressively prioritize Grade A/B and Venture/Education loans
- **Newbie Policy:** Apply lower initial limits for young renters/new employees to monitor behavior instead of rejection. This allows the bank to monitor repayment behavior before increasing exposure.

3. Early Warning System (EWS)

- **Automated Red-Flag Triggers:** Deploy system alerts for the "Default Formula": Gen Z + Low Income + Renting + DTI > 70% (Mandatory manual review).
- **Risk-Based Pricing Optimization:** Keep interest rates below the 15% "Breaking Point " This ensures borrowers remain solvent and improves long-term customer retention.